

(4.168)

$$\begin{aligned}
\vec{\gamma} &= -\frac{GM}{a^2} \left\{ \left[ 1 + (2\alpha + 3\alpha^2) \sin^2 B - 2\alpha^2 \sin^2 2B - 2\frac{h}{a} - 6\frac{h}{a} \alpha \sin^2 B + 3 \left( \frac{h}{a} \right)^2 \right. \right. \\
&\quad - 3J_2 \left( 1 + 4\alpha \sin^2 B - 4\frac{h}{a} \right) \left( \frac{3}{2} \sin^2 B - \frac{1}{2} - \frac{3}{2} \alpha \sin^2 2B \right) \\
&\quad - 5J_4 \left( \frac{5}{8} \sin^2 B - \frac{35}{32} \sin^2 2B + \frac{3}{8} \right) - m \left( 1 + \alpha - \alpha \sin^2 B + \frac{h}{a} \right) (1 - \sin^2 B + \alpha \sin^2 2B) \Big] \hat{e}_r \\
&\quad - \left[ J_2 \left( 1 + 4\alpha \sin^2 B - 4\frac{h}{a} \right) \left( \frac{3}{2} - 3\alpha + 6\alpha \sin^2 B \right) \sin 2B + J_4 \left( \frac{35}{4} \sin^2 B - \frac{15}{4} \right) \sin 2B \right. \\
&\quad \left. \left. + m \left( 1 + \alpha - \alpha \sin^2 B + \frac{h}{a} \right) \left( \frac{1}{2} - \alpha + 2\alpha \sin^2 B \right) \sin 2B \right] \hat{e}_\theta \right\} \\
&= -\frac{GM}{a^2} \left\{ \left[ 1 + \frac{3}{2} J_2 - \frac{15}{8} J_4 - m - m\alpha \right. \right. \\
&\quad + (2\alpha + 3\alpha^2 - \frac{9}{2} J_2 - 12J_2\alpha - \frac{25}{8} J_4 + m + m\alpha) \sin^2 B \\
&\quad + (-2\alpha^2 + 9\alpha J_2 + \frac{175}{32} J_4 - \frac{3}{4} m\alpha) \sin^2 2B \\
&\quad + [-6J_2 - 2 - m + (-6\alpha + 18J_2 + m) \sin^2 B] \frac{h}{a} + 3 \left( \frac{h}{a} \right)^2 \Big] \hat{e}_r \\
&\quad - \left[ \frac{3}{2} J_2 - 3J_2\alpha - \frac{15}{4} J_4 + \frac{1}{2} m - \frac{1}{2} m\alpha \right. \\
&\quad \left. \left. + \left( \frac{3}{2} m\alpha + 12J_2\alpha + \frac{35}{4} J_4 \right) \sin^2 B + \left( \frac{1}{2} m - 6J_2 \right) \frac{h}{a} \right] \sin 2B \hat{e}_\theta \right\}.
\end{aligned}$$

(4.169)

$$\begin{aligned}
\vec{\gamma} &= -\frac{GM}{a^2} \left\{ \left[ 1 + \frac{3}{2} \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) - \frac{15}{8} \left( -\frac{4}{5} \alpha^2 + \frac{4}{7} \alpha m \right) - m - m\alpha \right. \right. \\
&\quad + \left( 2\alpha + 3\alpha^2 - \frac{9}{2} \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) - 12 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) \alpha \right. \\
&\quad \left. \left. - \frac{25}{8} \left( -\frac{4}{5} \alpha^2 + \frac{4}{7} \alpha m \right) + m + m\alpha \right) \sin^2 B \right. \\
&\quad + \left( -2\alpha^2 + 9\alpha \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) + \frac{175}{32} \left( -\frac{4}{5} \alpha^2 + \frac{4}{7} \alpha m \right) - \frac{3}{4} m\alpha \right) \sin^2 2B \\
&\quad + \left[ -6 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) - 2 - m + (-6\alpha + 18 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) + m) \sin^2 B \right] \frac{h}{a} \\
&\quad \left. \left. + 3 \left( \frac{h}{a} \right)^2 \right] \hat{e}_r \right\}
\end{aligned}$$

$$\begin{aligned}
& - \left[ \frac{3}{2} \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) - 3 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) \alpha \right. \\
& - \frac{15}{4} \left( -\frac{4}{5} \alpha^2 + \frac{4}{7} \alpha m \right) + \frac{1}{2} m - \frac{1}{2} m \alpha \\
& + \left( \frac{3}{2} m \alpha + 12 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) \alpha + \frac{35}{4} \left( -\frac{4}{5} \alpha^2 + \frac{4}{7} \alpha m \right) \right) \sin^2 B \\
& \left. + \left( \frac{1}{2} m - 6 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) \right) \frac{h}{a} \right] \sin 2B \hat{e}_\theta \Big\} \\
= & - \frac{GM}{a^2} \left\{ \left[ 1 + (2\alpha + 3\alpha^2) \sin^2 B - 2\alpha^2 \sin^2 2B - 2\frac{h}{a} - 6\frac{h}{a} \alpha \sin^2 B + 3 \left( \frac{h}{a} \right)^2 \right. \right. \\
& - 3J_2 \left( 1 + 4\alpha \sin^2 B - 4\frac{h}{a} \right) \left( \frac{3}{2} \sin^2 B - \frac{1}{2} - \frac{3}{2} \alpha \sin^2 2B \right) \\
& - 5J_4 \left( \frac{5}{8} \sin^2 B - \frac{35}{32} \sin^2 2B + \frac{3}{8} \right) - m \left( 1 + \alpha - \alpha \sin^2 B + \frac{h}{a} \right) (1 - \sin^2 B + \alpha \sin^2 2B) \Big] \hat{e}_r \\
& - \left[ J_2 \left( 1 + 4\alpha \sin^2 B - 4\frac{h}{a} \right) \left( \frac{3}{2} - 3\alpha + 6\alpha \sin^2 B \right) \sin 2B + J_4 \left( \frac{35}{4} \sin^2 B - \frac{15}{4} \right) \sin 2B \right. \\
& \left. + m \left( 1 + \alpha - \alpha \sin^2 B + \frac{h}{a} \right) \left( \frac{1}{2} - \alpha + 2\alpha \sin^2 B \right) \sin 2B \right] \hat{e}_\theta \Big\} \\
= & - \frac{GM}{a^2} \left\{ \left[ 1 + \frac{3}{2} J_2 - \frac{15}{8} J_4 - m - m\alpha \right. \right. \\
& + (2\alpha + 3\alpha^2 - \frac{9}{2} J_2 - 12J_2\alpha - \frac{25}{8} J_4 + m + m\alpha) \sin^2 B \\
& + (-2\alpha^2 + 9\alpha J_2 + \frac{175}{32} J_4 - \frac{3}{4} m\alpha) \sin^2 2B \\
& + [-6J_2 - 2 - m + (-6\alpha + 18J_2 + m) \sin^2 B] \frac{h}{a} + 3 \left( \frac{h}{a} \right)^2 \Big] \hat{e}_r \\
& - \left[ \frac{3}{2} J_2 - 3J_2\alpha - \frac{15}{4} J_4 + \frac{1}{2} m - \frac{1}{2} m\alpha \right. \\
& \left. + \left( \frac{3}{2} m\alpha + 12J_2\alpha + \frac{35}{4} J_4 \right) \sin^2 B + \left( \frac{1}{2} m - 6J_2 \right) \frac{h}{a} \right] \sin 2B \hat{e}_\theta \Big\} . \\
= & - \frac{GM}{a^2} \left\{ \left[ 1 + \frac{3}{2} \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) - \frac{15}{8} \left( -\frac{4}{5} \alpha^2 + \frac{4}{7} \alpha m \right) - m - m\alpha \right. \right. \\
& + \left( 2\alpha + 3\alpha^2 - \frac{9}{2} \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) - 12 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) \alpha \right. \\
& \left. \left. - \frac{25}{8} \left( -\frac{4}{5} \alpha^2 + \frac{4}{7} \alpha m \right) + m + m\alpha \right) \sin^2 B \right. \\
& + \left( -2\alpha^2 + 9\alpha \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) + \frac{175}{32} \left( -\frac{4}{5} \alpha^2 + \frac{4}{7} \alpha m \right) - \frac{3}{4} m\alpha \right) \sin^2 2B \\
& + \left[ -6 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) - 2 - m + (-6\alpha + 18 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) + m) \sin^2 B \right] \frac{h}{a} \\
& \left. + 3 \left( \frac{h}{a} \right)^2 \right] \hat{e}_r \\
& + 3 \left( \frac{h}{a} \right)^2 \Big] \hat{e}_r
\end{aligned}$$

$$\begin{aligned}
& - \left[ \frac{3}{2} \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) - 3 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) \alpha \right. \\
& - \frac{15}{4} \left( -\frac{4}{5} \alpha^2 + \frac{4}{7} \alpha m \right) + \frac{1}{2} m - \frac{1}{2} m \alpha \\
& + \left( \frac{3}{2} m \alpha + 12 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) \alpha + \frac{35}{4} \left( -\frac{4}{5} \alpha^2 + \frac{4}{7} \alpha m \right) \right) \sin^2 B \\
& \left. + \left( \frac{1}{2} m - 6 \left( \frac{2}{3} \alpha - \frac{1}{3} m - \frac{1}{3} \alpha^2 + \frac{2}{21} \alpha m \right) \right) \frac{h}{a} \right] \sin 2B \hat{e}_\theta \Big\} \\
= & - \frac{GM}{a^2} \left\{ \left[ 1 + \alpha - \frac{1}{2} m - \frac{1}{2} \alpha^2 + \frac{1}{7} \alpha m + \frac{3}{2} \alpha^2 - \frac{15}{14} \alpha m - m - m \alpha \right. \right. \\
& + \left( 2\alpha + 3\alpha^2 - 3\alpha + \frac{3}{2} m + \frac{3}{2} \alpha^2 - \frac{3}{7} \alpha m - 8\alpha^2 + 4m\alpha + \frac{5}{2} \alpha^2 - \frac{25}{14} \alpha m + m + m\alpha \right) \sin^2 B \\
& + \left( -2\alpha^2 + 6\alpha^2 - 3\alpha m - \frac{35}{8} \alpha^2 + \frac{25}{8} \alpha m - \frac{3}{4} m\alpha \right) \sin^2 2B \\
& + \left[ -4\alpha + 2m + 2\alpha^2 - \frac{4}{7} \alpha m - 2 - m + \left( -6\alpha + 12\alpha - 6m - 6\alpha^2 + \frac{12}{7} \alpha m + m \right) \sin^2 B \right] \frac{h}{a} + 3 \left( \frac{h}{a} \right)^2 \Big] \hat{e}_r \\
& - \left[ \alpha - \frac{1}{2} m - \frac{1}{2} \alpha^2 + \frac{1}{7} \alpha m - 2\alpha^2 + m\alpha + 3\alpha^2 - \frac{15}{7} \alpha m + \frac{1}{2} m - \frac{1}{2} m\alpha \right. \\
& + \left( \frac{3}{2} m\alpha + 8\alpha^2 - 4m\alpha - 7\alpha^2 + 5\alpha m \right) \sin^2 B + \left( \frac{1}{2} m - 4\alpha + 2m + 2\alpha^2 + \frac{4}{7} \alpha m \right) \frac{h}{a} \Big] \sin 2B \hat{e}_\theta \Big\} \\
= & - \frac{GM}{a^2} \left\{ \left[ 1 + \alpha - \frac{1}{2} m - m - \frac{1}{2} \alpha^2 + \frac{3}{2} \alpha^2 + \frac{1}{7} \alpha m - \frac{15}{14} \alpha m - m\alpha \right. \right. \\
& + \left( 2\alpha - 3\alpha + \frac{3}{2} m + m + 3\alpha^2 - 8\alpha^2 + \frac{5}{2} \alpha^2 + \frac{3}{2} \alpha^2 - \frac{3}{7} \alpha m + 4m\alpha - \frac{25}{14} \alpha m + m\alpha \right) \sin^2 B \\
& + \left( -2\alpha^2 + 6\alpha^2 - \frac{35}{8} \alpha^2 - 3\alpha m + \frac{25}{8} \alpha m - \frac{3}{4} m\alpha \right) \sin^2 2B \\
& + \left[ -2 - 4\alpha - m + 2m + 2\alpha^2 - \frac{4}{7} \alpha m + \left( -6\alpha + 12\alpha - 6m + m - 6\alpha^2 + \frac{12}{7} \alpha m \right) \sin^2 B \right] \frac{h}{a} + 3 \left( \frac{h}{a} \right)^2 \Big] \hat{e}_r \\
& - \left[ \alpha - \frac{1}{2} m + \frac{1}{2} m - \frac{1}{2} \alpha^2 + 3\alpha^2 - 2\alpha^2 + \frac{1}{7} \alpha m + m\alpha - \frac{15}{7} \alpha m - \frac{1}{2} m\alpha \right. \\
& + \left( 8\alpha^2 - 7\alpha^2 + \frac{3}{2} m\alpha - 4m\alpha + 5\alpha m \right) \sin^2 B + \left( -4\alpha + \frac{1}{2} m + 2m + 2\alpha^2 + \frac{4}{7} \alpha m \right) \frac{h}{a} \Big] \sin 2B \hat{e}_\theta \Big\} \\
= & - \frac{GM}{a^2} \left\{ \left[ 1 + \alpha - \frac{3}{2} m + \alpha^2 - \frac{27}{14} \alpha m + \left( -\alpha + \frac{5}{2} m - \alpha^2 + \frac{39}{14} \alpha m \right) \sin^2 B + \left( -\frac{3}{8} \alpha^2 - \frac{5}{8} \alpha m \right) \sin^2 2B \right. \right. \\
& + \left[ -2 - 4\alpha + m + \left( 6\alpha - 5m \right) \sin^2 B \right] \frac{h}{a} + 3 \left( \frac{h}{a} \right)^2 \Big] \hat{e}_r \\
& - \left[ \alpha + \frac{1}{2} \alpha^2 - \frac{3}{2} m\alpha + \left( \alpha^2 + \frac{5}{2} m\alpha \right) \sin^2 B + \left( -4\alpha + \frac{5}{2} m \right) \frac{h}{a} \right] \sin 2B \hat{e}_\theta \Big\} \\
= & - \frac{GM}{a^2} \left\{ \left[ 1 + \alpha - \frac{3}{2} m + \alpha^2 - \frac{27}{14} \alpha m + \left( -\alpha + \frac{5}{2} m - \alpha^2 + \frac{39}{14} \alpha m \right) \sin^2 B \right. \right.
\end{aligned}$$

$$\begin{aligned}
& + \left( -\frac{3}{8}\alpha^2 - \frac{5}{8}\alpha m \right) \sin^2 2B + [-2 - 4\alpha + m + (6\alpha - 5m) \sin^2 B] \frac{h}{a} + 3 \left( \frac{h}{a} \right)^2 \Big] \hat{e}_r \\
& - \left[ \alpha + \frac{1}{2}\alpha^2 - \frac{3}{2}m\alpha + \left( \alpha^2 + \frac{5}{2}m\alpha \right) \sin^2 B + \left( -4\alpha + \frac{5}{2}m \right) \frac{h}{a} \right] \sin 2B \hat{e}_\theta \Big\}.
\end{aligned}$$

**(4.170)**

$$\begin{aligned}
\vec{\gamma} &= -\gamma_0 \left\{ \left[ 1 - \frac{1}{2}\alpha^2 \sin^2 2B \right. \right. \\
&+ \left[ 1 + \alpha - \frac{3}{2}m + \left( -\alpha + \frac{5}{2}m \right) \sin^2 B \right]^{-1} [-2 - 4\alpha + m + (6\alpha - 5m) \sin^2 B] \frac{h}{a} + 3 \left( \frac{h}{a} \right)^2 \Big] \hat{e}_r \\
&- \left[ 1 + \alpha - \frac{3}{2}m + \left( -\alpha + \frac{5}{2}m \right) \sin^2 B \right]^{-1} \left[ \alpha + \frac{1}{2}\alpha^2 - \frac{3}{2}m\alpha + \left( \alpha^2 + \frac{5}{2}m\alpha \right) \sin^2 B + \left( -4\alpha + \frac{5}{2}m \right) \frac{h}{a} \right] \sin 2B \hat{e}_\theta \Big\} \\
&= -\gamma_0 \left\{ \left[ 1 - \frac{1}{2}\alpha^2 \sin^2 2B \right. \right. \\
&+ \left[ 1 - \alpha + \frac{3}{2}m - \left( -\alpha + \frac{5}{2}m \right) \sin^2 B \right] [-2 - 4\alpha + m + (6\alpha - 5m) \sin^2 B] \frac{h}{a} + 3 \left( \frac{h}{a} \right)^2 \Big] \hat{e}_r \\
&- \left[ 1 - \alpha + \frac{3}{2}m - \left( -\alpha + \frac{5}{2}m \right) \sin^2 B \right] \left[ \alpha + \frac{1}{2}\alpha^2 - \frac{3}{2}m\alpha + \left( \alpha^2 + \frac{5}{2}m\alpha \right) \sin^2 B + \left( -4\alpha + \frac{5}{2}m \right) \frac{h}{a} \right] \sin 2B \hat{e}_\theta \Big\} \\
&= -\gamma_0 \left\{ \left[ 1 - \frac{1}{2}\alpha^2 \sin^2 2B \right. \right. \\
&+ [2\alpha - 3m + 2 \left( -\alpha + \frac{5}{2}m \right) \sin^2 B - 2 - 4\alpha + m + (6\alpha - 5m) \sin^2 B] \frac{h}{a} + 3 \left( \frac{h}{a} \right)^2 \Big] \hat{e}_r \\
&- \left[ -\alpha^2 + \frac{3}{2}\alpha m - \alpha \left( -\alpha + \frac{5}{2}m \right) \sin^2 B + \alpha + \frac{1}{2}\alpha^2 - \frac{3}{2}m\alpha + \left( \alpha^2 + \frac{5}{2}m\alpha \right) \sin^2 B \right. \\
&+ \left. \left( -4\alpha + \frac{5}{2}m \right) \frac{h}{a} \right] \sin 2B \hat{e}_\theta \Big\} \\
&= -\gamma_0 \left\{ \left[ 1 - \frac{1}{2}\alpha^2 \sin^2 2B - 2[1 + \alpha + m - 2\alpha \sin^2 B] \frac{h}{a} + 3 \left( \frac{h}{a} \right)^2 \right] \hat{e}_r \right. \\
&- \left. \left[ \alpha - \frac{1}{2}\alpha^2 + 2\alpha^2 \sin^2 B + \left( -4\alpha + \frac{5}{2}m \right) \frac{h}{a} \right] \sin 2B \hat{e}_\theta \right\}. \tag{1}
\end{aligned}$$